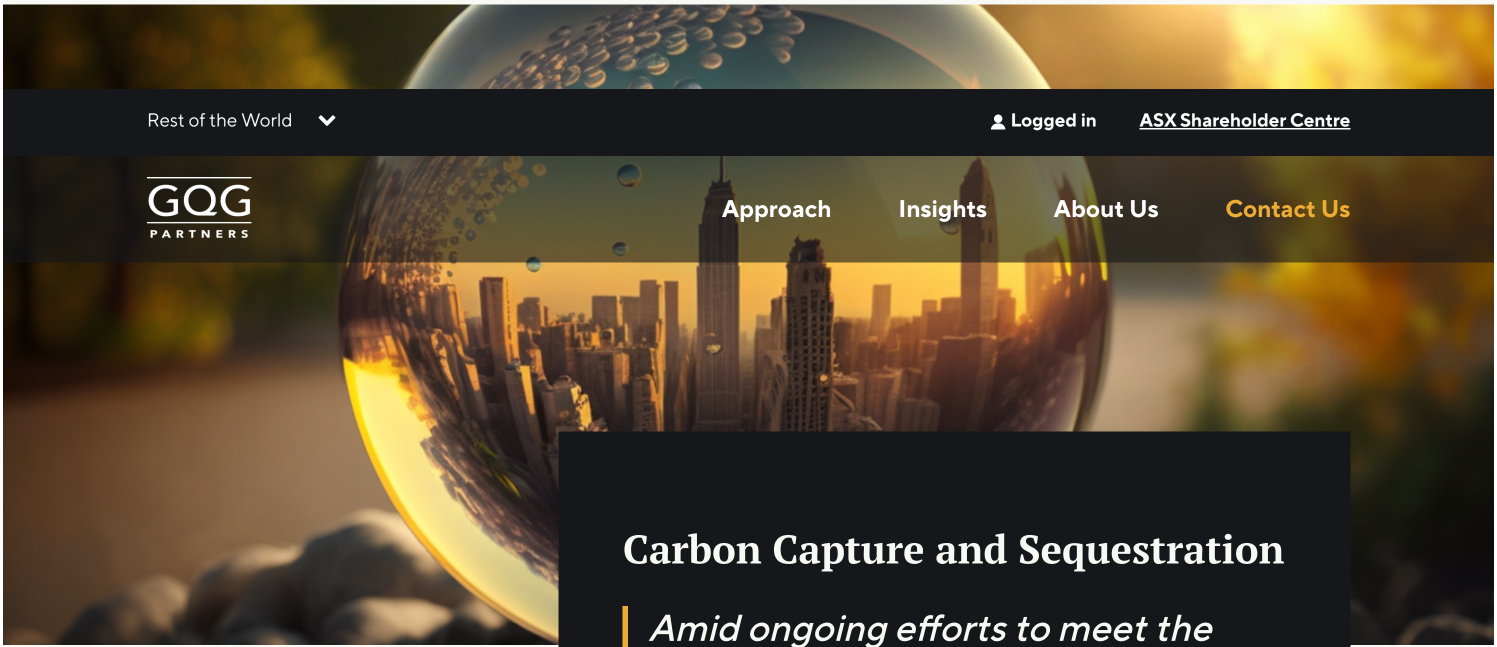


Carbon Capture and Sequestration

Amid ongoing efforts to meet the targets of the Paris Agreement and limit the global temperature rise by 2050

June 9, 2023

GQG Research

Key Takeaways

- Carbon capture and sequestration (CCS) is emerging as a crucial tool in mitigating climate change, offering both a solution to emissions reduction and a new business model for companies in the form of low-carbon fuels, CO2 feedstocks, and sequestration credits
- Integrated oil companies (IOCs), particularly ExxonMobil and Occidental Petroleum, are leading in CCS efforts, leveraging their existing infrastructure and expertise to capture and store more carbon than other oil majors
- Government support, such as tax credits and infrastructure investment, is boosting the viability and profitability of CCS, positioning it as a promising investment opportunity and a key component of the global energy transition

NUANCED ISSUES CALL FOR NOVEL SOLUTIONS

Amid ongoing efforts to meet the targets of the Paris Agreement and limit the global temperature rise to 2°C by 2050, both governments and industries face a tough challenge: addressing emissions mitigation while accounting for the increased demand for fossil fuels. In our view, carbon capture will be an important comparative advantage as governments, investors, and other stakeholders look for leaders in climate change mitigation.

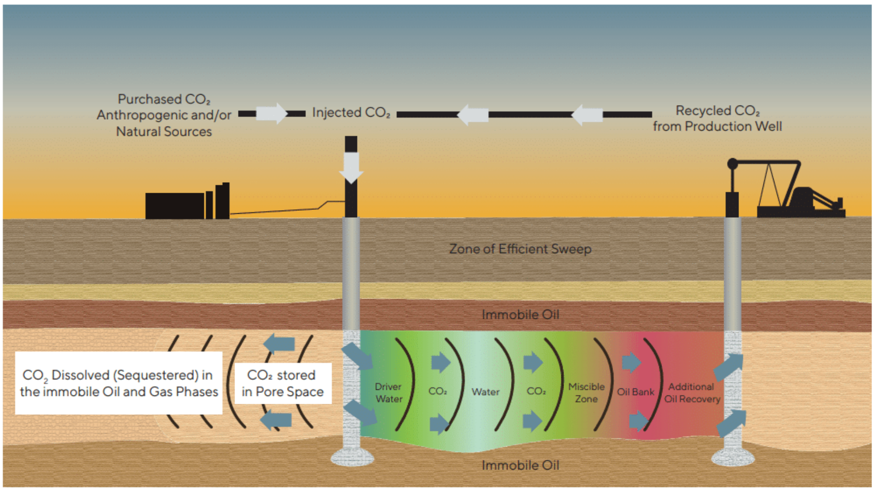
While investment in carbon capture has largely been viewed as a moral quandary rather than an investment opportunity, the dynamics are changing. In fact, carbon capture has acted as a hedge to stranded asset risk for years, and now provides a new model for revenue growth in the form of low-carbon fuels and chemicals, CO_2 based feedstocks, and sequestration credits.

As it stands, we believe developed-market upstream producers are in the best position to take advantage of carbon capture and sequestration (CCS) and contribute to the global energy transition. Let's delve deeper into the carbon capture efforts of integrated oil companies (IOCs), specifically ExxonMobil (XOM) and Occidental Petroleum (OXY), which are leaders in the space despite criticism that they are not doing enough to reduce their emissions.

CARBON CAPTURE: MIT CLIMATE PORTAL

Carbon capture technology was patented in 1952 by the Atlantic Refining Company and became commercialized in the 1970s with a process known as enhanced oil recovery (EOR). The EOR process involves the injection and recycling of CO_2 from a point-source through a reservoir to recover oil when conventional drilling is no longer viable. Today, this process is used to produce what the industry refers to as "low" or "zero-carbon" fuel, as the captured CO_2 is recycled through a closedloop system and, at times, partially stored in subsurface reservoirs.

EXHIBIT 1: HOW CARBON CAPTURE WORKS



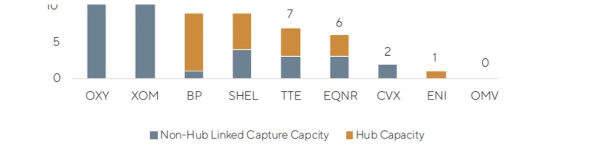
Source: National Energy Technology Laboratory, U.S. Department of Energy

CCS consists of three main components that are also utilized in the EOR process: capture, transportation, and storage of CO_2. Since carbon capture for EOR was first utilized in the Permian Basin in West Texas, traditionally US based oil majors have had an advantage in the development and expansion of this technology. Companies like Occidental and ExxonMobil now operate some of the largest EOR projects globally and maintain the infrastructure of pipelines needed to transport and store carbon. As a result, Occidental and ExxonMobil capture more carbon than any other oil majors, as noted in the chart below.

EXHIBIT 2: OPERATIONAL AND ANDVANCED DEVELOPMENT CCS CAPACITY

AN ALTERNATIVE – DIRECT AIR CAPTURE

Another CCS solution being explored by Occidental and ExxonMobil is that of direct air



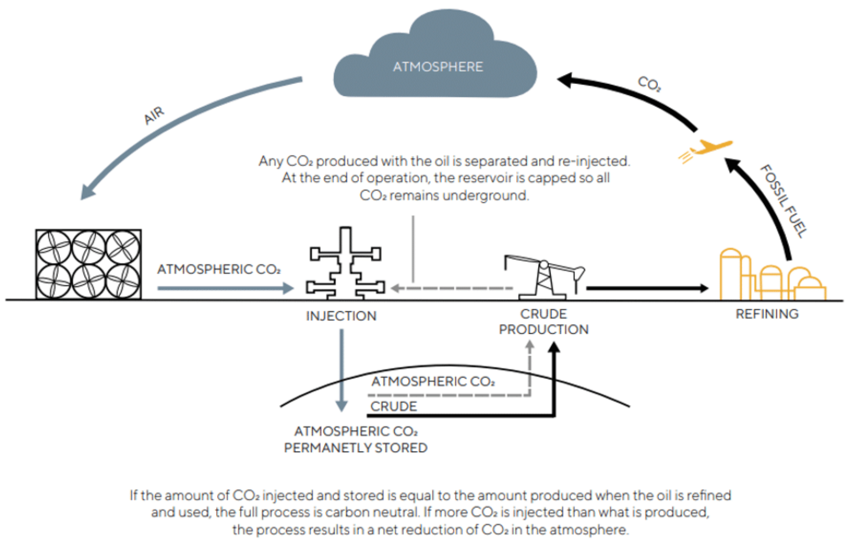
Source: Jefferies estimates, Company data, Bloomberg, CCS Institute.

capture (DAC), which is technologically mature and reaching commercial viability. DAC technologies remove CO_2 from the air through the use of chemical sorbents or solvents.

Solvent plants use liquid solvent to separate and absorb CO_2 as air passes through the system, resulting in the formation of a stable solid mineral (CO_2 and Ca).[1,2]

Heat is then used to re-gasify the CO_2 for EOR or sequestration. Sorbent plants use solid sorbent as a filter to bind with the captured CO_2 and then apply low-temperature heat to release the CO_2.[1,2]

EXHIBIT 3: HOW DIRECT AIR CAPTURE WORKS



Source: CarbonEngineering, company that partners with OXY & XOM - Direct Air Capture and Storage of CO_2 | Carbon Engineering.

When compared to pointsource capture, DAC is more energy intensive and expensive as atmospheric CO_2 is less concentrated than flue gas from a power station or a cement plant. This leads to higher energy demand and costs relative to other CO_2 capture technologies and applications. However, the deployment of DAC plants will be key to reducing absolute global emissions.

SEQUESTRATION HUBS

The development of sequestration hubs, which aggregate emissions from multiple sources, is a major opportunity for IOCs. It is the most efficient way to decarbonize industrial clusters and achieve economies of scale. Shared storage sites also reduce the costs of drilling wells, managing injections, and monitoring sites for leakage. By minimizing transport and storage costs through the use of hub projects, we believe CCS can become more viable. ExxonMobil is leading the charge in hub capacity, with its Houston Hub projected to capture and store ~50 million tons per annum (mtpa) of CO_2 by 2030. This project is a collaboration between ExxonMobil, Chevron, Dow, Linde, Valero, and other industrial and chemical companies.

EXHIBIT 4: IOC ADVANCED DEVELOPMENT AND UNDER CONSTRUCTION CCS HUBS[3]

UC & AD Projects	Gross Capacity (mtpa)	Start Date	IOCs with Equity Stake in Hub	IOCs with Connected Facilities
Northern Lights Phase 1	1.5	2024	EQNR, SHEL, TTE	-
Porthos Phase 1	2.5	2024	-	XOM, SHEL
Acorn CCS Phase 1 (Gas Processing)	0.3	2025	SHEL	SHEL, XOM
HyNet North West Phase 1	1.0	2025	ENI	-
West Bay Sequestration Hub	6.0	2025	OXY	-
Polaris CCS Phase 1 (Canada)	0.7	2026	SHEL	SHEL
Antwerp@C Phase 1	1.0	2026	-	XOM, TTE, OMV
Aramis CCS Phase 1	5.0	2026	TTE, SHEL	TTE, SHEL, EQNR
East Coast Cluster Phase 1	17.0	2026	BP, EQNR, TTE, SHEL	BP, EQNR
NEP: Net Zero Teeside Phase 1	4.0	2026	BP, EQNR	BP, EQNR
NEP: Zero Carbon Humber Phase 1	9.0	2026	-	EQNR
Houston CCS Innovation Zone Phase 1	10.0	2027	XOM	CVX, SHEL
Total	58.0			

Source: Jefferies estimates, Company data, Bloomberg, CCS Institute.

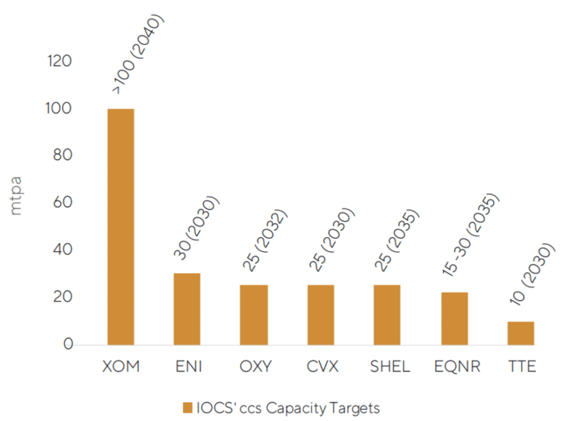
CAPACITY AND GROWTH

Based on the sixth assessment report by the Intergovernmental Panel on Climate Change (IPCC AR6), achieving net zero by 2050 requires the capture of ~10 gigatons (GT) of CO_2per year, which represents nearly a quarter of global annual emissions. Current global capture capacity stands at 44 mtpa[4] – the equivalent of emissions from over 8 million passenger cars.[5] Although the world is far short of capturing 10 GT annually, in 2022 alone, the total capture capacity of commercial CCS projects (operational and in-development) grew to 244 mtpa[6] – a 44 per cent increase year-over-year.

We learned through one of our engagements with Occidental that the total cost of a DAC plant stands at ~$1.4 billion, of which Occidental will contribute $275 million for its first DAC plant. Although guidance on further allocations towards direct air capture has not been disclosed, the company recently outlined its ambition to have 100 DAC plants online by 2035 and will invest between $100 million and $500 million in its low carbon ventures segment. Occidental was also confident that its Low Carbon Ventures unit will eventually eclipse OxyChem sales, which represented ~19 per cent of Occidental's total revenue in 2022."

As illustrated in Exhibit 2, Occidental and ExxonMobil have the largest project portfolios due to legacy EOR applications in the Permian Basin. Exhibit 5 illustrates the capacity targets for leading IOCs, with ExxonMobil appearing the most ambitious. However, as Occidental currently captures ~20 mtpa, the company appears best positioned to achieve its capture targets. ExxonMobil's capacity targets are based on projected DAC deployment and hub capacity, further detailed below, which are still under development. Despite the comparative advantage of US IOCs, peers across the pond are utilizing carbon capture technologies as well. BP and Shell lead the Europeans in current capture capacity, while TotalEnergies and ENI SpA have significant room for growth. In mid-2022, TotalEnergies signed its first commercial agreement to permanently store

EXHIBIT 5: IOCS' CCS CAPACITY TARGETS



Source: Jefferies estimates, Company data, Bloomberg, CCS Institute.

~800,000 tons of CO_2 per year beginning in 2025. This is just one of multiple advanced market commitments that are now being announced by industry groups and investors.

WHAT'S MOVING THE NEEDLE?

Policymakers have now come to the conclusion that incentivizing the storage of carbon is a key lever in driving global decarbonization. This presents further opportunities for legacy operators in the carbon capture space, as it extends the useful life of their assets and provides potentially higher revenue streams for existing storage infrastructure. In the US, government support has come from the 45Q tax credit — which provides $35 per ton for CO_2 used in EOR and $50/t for permanent storage of CO_2 up to 2026.[7] The CCUS Tax Credit Amendments Act of 2021 proposed an increase in the value of the tax credit from $35/t to $75/t for EOR facilities and from $50/t to $120/t for facilities that sequester CO_2. [8] Additionally, with the passage of the Infrastructure Investment and Jobs Act, Congress provided the Department of Energy with $3.5 billion for its Regional Direct Air Capture Hubs program. This is to be spent over the next five years across four DAC hub projects. Eligibility is tied to a project's ability to capture 1 million tons annually upon completion, giving larger IOCs an upper hand due to the balance sheet needed to finance these developments.[9]

Following the passage of the Inflation Reduction Act (IRA), incentives for deploying carbon capture are now clear and the investment case is stronger than ever, in our opinion. The IRA increased 45Q tax credits to $85/MT for industrial carbon capture, utilization and storage (CCUS) and $180/MT for DAC (non-EOR). Most notably, 45Q tax credits generated in the first five years of operation will be paid upfront.[10] Outside of the US, an agreement was reached at COP26 to develop a global carbon market with standardized rules between governments and corporations.[11] This agreement formalized the creation of a centralized system open to the public and private sectors, and enhanced the stringency of rules governing the trade of credits between countries.[12] It will have direct, immediate impacts to ongoing offset programs, such as CORSIA, which manages the global offsetting system for airlines. While the rules will be stricter, the agreement provides greater clarity to enable greater participation in the market.

OCCIDENTAL PETROLEUM'S CCS PLANTS

We believe Occidental is well positioned to achieve their capture targets versus peers. Currently, Occidental has the greatest operational capacity for CCS projects through its legacy businesses in the Permian Basin. The company is also building experience through pilot projects with industry partners. Additionally, the company has an advanced development DAC project, aptly named DAC One. The project is expected to come online near the end of 2024 and should capture 0.5 mtpa, representing nearly $90 million of 45Q credit generation annually at $180/t. For any emissions that Occidental offtakes and stores, the point-source emitter will capture the $85/t credit and pay the company a portion for the transportation and sequestration services. As these credits are directly payable for the first five years, Occidental has the greatest near-term revenue opportunity from credit generation.

CONCLUSION

Carbon capture is, and will likely continue to be, an important solution as governments, investors, and other stakeholders look for leaders in climate change mitigation. The uses of CCS technology now extend beyond the oil and gas industry, with applications in steel, cement, and chemical production. This demonstrates the viability of and efficacy in CCS technology.

As the pipeline of CCS projects continues to grow, an important question to consider is whether we have the geological space to sequester the millions of tons of CO_2. Fortunately, the answer to that question is yes. The IPCC estimates that technical geological storage capacity stands at nearly 1,000 Gigatons of CO_2. To put it in perspective, this is more than the CO_2 storage needed through 2100 to limit global warming to 1.5°C. However, in order to reach this goal by 2050, greater deployment of CCS technologies will be needed as well as supportive regulatory frameworks from government institutions.

Legacy oil and gas producers already have the patents and partnerships needed to deploy CCS, as well as healthy balance sheets to fund these projects. Perhaps most importantly, major IOCs need these developments to succeed in order to show how they are addressing the externalities associated with their operations. Companies such as ExxonMobil and Occidental have made clear their commitment to expanding the use of CCS technology, which represents not only a solution to lowering global emissions but also a valuable business opportunity.

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